Demographics and Housing Booms

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Abstract

We show that the age structure of the population systematically shapes housing-boom vulnerability. Across OECD countries and U.S. counties, a one-unit increase in the young-to-old ratio raises the probability of a large housing boom by 18.6 and 4.3 percentage points, respectively. In mortgage application data, young applicants increase their application intensity 4.3 to 6.6 percent more than older applicants during local booms, and after an identified credit-supply easing, house prices rise more, and more persistently, in younger areas. We then show that a life-cycle housing model accounts for the larger initial price response in younger economies, because young households are concentrated at the margin between owning and renting and respond most to credit easing.

JEL codes: G51; E52; R31; J11

Keywords: Housing Booms; Age Structure; Mortgage Credit Supply; Life-Cycle Model

1 Introduction

The relationship between demographic structure and housing markets has long been a central topic in macroeconomics and housing economics. A substantial body of existing research has shown that the age composition of households is an important determinant of the level and long-run trend of housing demand. For example, it is well known that demand for housing space and homeownership rises sharply at particular stages of the life cycle, while housing consumption and asset composition change after retirement (Mankiw and Weil, 1989; Bakshi and Chen, 1994; Takáts, 2012; Nishimura and Takáts, 2012).

However, much less is known about whether demographic structure affects not only these long-run trends, but also the likelihood and magnitude of house price booms. Since young households are more likely to be located at the margin between homeownership and renting, actual home purchases may increase more strongly in response to the same demand shock. In other words, demographic composition may be an important state variable governing both the likelihood and the magnitude of house price booms. Understanding this relationship is important for identifying the sources of housing-market fluctuations and for designing the timing and intensity of policy responses. If demographic structure amplifies or dampens the effects of a given shock, demographic composition can be interpreted as a key state variable for predicting the risk of house price booms and financial instability.

This paper asks whether the age structure of the population governs a location's vulnerability to housing booms. We find that younger locations are more vulnerable, and we interpret this pattern through a life-cycle housing-demand mechanism. The value of owning rather than renting is largest for households with the longest horizon: owning hedges future rent risk (Sinai and Souleles, 2005), and the present value of the flow benefits of ownership rises with the years spent in the home. Younger households therefore have the most to gain from owning, and they are the cohort concentrated at the first-time-buyer margin (Ortalo-Magne and Rady, 2006), so their purchase decision responds most to a credit supply shock.

Our empirical analysis begins by documenting the relationship between age structure and the incidence of housing booms. We define a large housing boom as an episode in which the cyclical

component of real house prices is more than one standard deviation above its region-specific trend. Using both a panel of 29 OECD countries from 1970 to 2021 and a U.S. county-year panel from 1975 to 2024, we show that locations with a higher ratio of young adults to older adults are more likely to experience large upward deviations of house prices from trend.

In the linear probability model using the country panel, a one-unit increase in the young-to-old ratio, defined as the population aged 20–39 relative to the population aged 50–69, is associated with a statistically significant 18.6 percentage point increase in the probability of a large housing boom. The corresponding estimate from the U.S. county panel implies a 4.3 percentage point increase in the probability of a housing boom. Thus, across different datasets and specifications, the evidence indicates that a higher young-to-old ratio is associated with a greater incidence of housing booms.[1]

We next ask whether the greater incidence of booms in younger regions reflects a rise in home-buying by young households. Using the application data from the Home Mortgage Disclosure Act (HMDA), we examine whether young applicants increase home-purchase activity more strongly when local house prices are in a boom, thereby contributing disproportionately to mortgage-market activity. The results show that, in counties experiencing a housing boom, the application intensity of young applicants is 4.3–6.6 percent higher when normalized by the boom-period average of the older comparison group. These findings are robust across alternative definitions of age groups, application-based outcome variables, and more stringent boom thresholds.

In the second part of the paper, we ask whether a standard life-cycle housing model can rationalize these facts and reproduce their dynamics. The model is deliberately conventional: an annual overlapping-generations economy over ages 25 to 84, with a discrete own-rent choice, a fixed-rate mortgage subject to a loan-to-value limit, a hump-shaped life-cycle income profile, and a fixed stock of owner-occupied housing, following Ortalo-Magne and Rady (2006), Berger et al. (2018), and Kaplan et al. (2020). Holding preferences, the interest rate, the mortgage contract, and the calibration fixed, we vary only the age structure, and we feed the economy the same identified

[1] This relationship does not fully rule out the possibility of reverse causality, whereby young households move into regions experiencing rapid house-price growth. However, that the results appear not only in U.S. county-level data, where migration can operate relatively quickly, but also in country-level panels is more directly consistent with a life-cycle housing-demand mechanism than with a migration explanation.

credit shock we estimate in the data. This isolates the demographic channel and places the model on the same footing as the reduced-form evidence.

The shock is a compression in the mortgage spread, the channel through which the Large-Scale Asset Purchase (LSAP) easing operates. Cheaper credit lowers the user cost of owning most for households near the owner-renter margin, who are disproportionately young, so the same aggregate easing has an age-graded incidence. Re-clearing the fixed housing stock, a younger economy delivers a larger price response, about 1.3% at the peak against 1.2% in an old economy. In the baseline, where households expect no further price change, the model reproduces the untargeted impact age gradient of the identified response in data, but generates none of its persistence. Once we introduce adaptive house-price expectations, the response persists and the model reproduces the dynamic gradient of the empirical finding, which we read as suggestive evidence that an expectations channel matters for the dynamics. The model thus supplies the life-cycle mechanism behind the boom-incidence and mortgage-participation facts.

Related literature. This paper is related to three strands of literature. First, the paper contributes to the literature on demographics and housing markets. From a life-cycle perspective, age structure shapes household formation, saving behavior, portfolio allocation, and the demand for owner-occupied housing (Bakshi and Chen, 1994). The classic "baby boom" hypothesis of Mankiw and Weil (1989) argues that changes in cohort size generate large shifts in housing demand and house prices. Subsequent work shows that population aging can affect house prices and asset valuations over long horizons (Takáts, 2010, 2012; Nishimura and Takáts, 2012; Gevorgyan, 2019). This literature establishes that demographics are a determinant of long-run housing demand and valuation. Our paper shifts the focus from long-run price levels to boom vulnerability. We show that demographic composition also predicts the likelihood of large upward deviations of house prices from trend.

Second, the paper contributes to the literature on housing booms, credit constraints, and macro-financial amplification. A large literature studies housing booms as unusually large deviations of real house prices from trend and emphasizes the macro-financial risks associated with asset-price booms (Borio and Lowe, 2002; Detken and Smets, 2004; Goodhart and Hofmann, 2008; Jordà

et al., 2015). Closely related work studies how credit conditions, household leverage, housing finance, and beliefs amplify housing boom-bust cycles (Mian and Sufi, 2011; Favilukis et al., 2017; Justiniano et al., 2019; Kaplan et al., 2020). Our analysis builds on this literature by asking why similar housing-market shocks generate larger booms in some locations than in others. While urban research emphasizes housing-supply elasticity as a key source of local boom sensitivity (Glaeser et al., 2008; Saiz, 2010a), we highlight a complementary demand-side state variable: the local young-to-old population ratio. To distinguish this demographic channel from variation in local supply conditions, our empirical specifications explicitly control for housing-supply elasticity, allowing the demographic gradient to be identified conditional on differences in supply responsiveness.

Third, the paper contributes to quantitative life-cycle and OLG models of housing demand. Using a deliberately parsimonious model, we isolate the role of age structure in shaping the house-price response to a credit shock. We also provide suggestive evidence that, combined with adaptive house-price expectations, the credit shock generates a persistent house-price response in younger societies, consistent with Kaplan et al. (2020).

Several of these models contain rich life-cycle structure in which younger households respond differently to shocks. In Kaplan et al. (2020) and Favilukis et al. (2017), however, overlapping generations serve to trace how a single economy's prices respond over time to an aggregate driver, shifting beliefs about housing demand and a relaxation of housing-finance constraints respectively, rather than as an object of comparison across economies. Our contribution is instead to treat the age composition of the population as a cross-regional state variable: holding the shock fixed, younger economies place more households at the owner-renter margin and therefore transmit a common shock more strongly into prices. We are closest to Ortalo-Magne and Rady (2006), who show that credit-constrained young first-time buyers drive housing dynamics through the property ladder, but we recast their within-market logic as a comparative-statics statement about which regions are most boom-prone, distinct from work that uses life-cycle structure to explain the consumption response to house prices (Berger et al., 2018) or long-run demographic trends in price levels (Mankiw and Weil, 1989; Takáts, 2012).

The rest of the paper proceeds as follows. Section 2 presents the empirical evidence on demographics and large housing booms. Section 3 develops the housing OLG model and uses it to assess whether the empirical pattern can be rationalized by a life-cycle housing-demand mechanism. Section 4 concludes.

2 Effects of Demographics on Housing Booms

2.1 Data

Our empirical analysis combines several datasets at both the country and regional levels. For the cross-country analysis, we use OECD house price indicators, OECD population data by age group, and real GDP per capita for 29 countries over the period 1970–2021. For the U.S. county analysis, we use annual county house price indexes from the Federal Housing Finance Agency (FHFA) over 1975–2024, together with county population data and housing supply elasticity estimates from Saiz (2010a). The number of county-year observations differs across specifications because housing-price availability and the availability of the supply-elasticity control are not identical across counties.

Table 1: Summary of Datasets

Analysis	Source	Unit	Sample	Variables
Cross-country	OECD	Country × year	1970–2021	Real house prices, real GDP per capita, population by age group
U.S. county	FHFA, U.S. Census	County × year	1975–2024	Real house prices, population by age group, supply elasticity

Notes: For the U.S. regional analysis, housing supply elasticity is from Saiz (2010a).

2.2 Large Housing Booms

We define a large housing boom as an episode in which real house prices rise substantially above their long-run trend. This definition builds on the literature on housing booms and bubbles. For example, Borio and Lowe (2002), Detken and Smets (2004), and Goodhart and Hofmann (2008) define housing booms as periods in which real house prices rise sufficiently above their HP-filtered trend under relatively high smoothing parameters. Jordà et al. (2015) identify housing

bubble episodes using two conditions: the cyclical component of real house prices must exceed one standard deviation, and house prices must subsequently decline by at least 15 percent within three years.

In this study, we follow Jordà et al. (2015) in using the HP filter (Hodrick and Prescott, 1997) to extract the cyclical component of real house prices, but we omit the subsequent price-decline condition. By dropping this ex-post crash requirement, we retain episodes in which house prices rise sharply above trend, regardless of whether a large decline materializes within a specific window.

This modification is motivated by our different research objective. Whereas the definition in Jordà et al. (2015) is designed to identify boom-bust episodes and study the macro-financial consequences of bursting housing bubbles, our focus is on the incidence of large upward house-price deviations associated with demographic structure. Requiring an ex-post price collapse would condition the boom definition on a subsequent outcome that may depend on factors outside the demographic mechanism, such as credit conditions, policy responses, housing supply, or monetary tightening. We therefore retain the price-elevation criterion but omit the subsequent crash requirement, so that our boom indicator captures large upward deviations in real house prices rather than realized boom-bust episodes.

We decompose real house prices into trend and cyclical components using the HP filter:

$$p_{i,t} = \tau_{i,t} + c_{i,t}, \tag{1}$$

where $p_{i,t}$ denotes the log real house price in location i at time t, $\tau_{i,t}$ is the trend component, and $c_{i,t}$ is the cyclical component. Because both the country and county panels are annual, we set the HP smoothing parameter to $\lambda = 100$.

We then define the large housing boom indicator as

$$\text{LHB}_{i,t} = \begin{cases} 1, & \text{if } c_{i,t} > \sigma_{c_i}, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where σ_{c_i} is the standard deviation of $c_{i,t}$ for location i over the sample period. Thus, a large housing boom is identified whenever the cyclical component of real house prices exceeds one standard deviation of the location-specific cyclical component. Because the HP trend and σ_{c_i} are constructed from the available sample, the indicator should be interpreted as an ex-post classification of large upward deviations rather than as a real-time forecast signal.

2.3 Young-to-Old Population Ratio

The literature does not adopt a single age cutoff for distinguishing young, middle-aged, and old cohorts. The relevant boundaries vary across studies depending on the question of interest. In the housing literature, age profiles are typically motivated by life-cycle patterns in household formation, housing demand, and asset accumulation. The demographic literature similarly uses alternative cohort definitions depending on the mechanism under consideration.

We define the young cohort as individuals aged 20–39 and the older cohort as individuals aged 50–69. The numerator is intended to capture younger adults who are entering the housing market and forming housing demand, whereas the denominator captures older households who are, on average, more likely to have accumulated housing wealth and to adjust housing consumption or rebalance their portfolios. Accordingly, our measure should be interpreted not as a conventional old-age dependency ratio, but as a cohort-balance measure designed to capture life-cycle patterns in housing demand and asset holdings.

We define the young-to-old population ratio as

$$\mathrm{YO}_{i,t} \equiv \frac{\mathrm{Population}_{i,t}^{(20\text{–}39)}}{\mathrm{Population}_{i,t}^{(50\text{–}69)}}, \tag{3}$$

where a higher value of $\mathrm{YO}_{i,t}$ indicates a relatively larger young cohort compared with the older cohort. In the Appendix, we examine the robustness of the results to alternative age-group definitions.

2.4 Descriptive Evidence

Before turning to regression analysis, we present descriptive evidence on the relationship between demographic structure and large housing booms. Figure 1 plots the probability of a large housing boom against the young-to-old population ratio in the country panel and the U.S. county panel. Panel (a) displays the country-level relationship, while Panel (b) shows the analogous relationship at the U.S. county level.[2]

The country-level plot reveals a clear positive association: countries with a higher young-to-old ratio tend to experience large housing booms more frequently. The U.S. county-level plot shows a similar positive relationship, although the association is weaker than in the country panel. This difference is not surprising given the finer geographic resolution and greater heterogeneity of local housing markets. These patterns are descriptive, but they are consistent with the hypothesis that younger demographic structures are associated with more frequent large housing booms and motivate the regression analysis that follows.

2.5 Demographics and Large Housing Booms

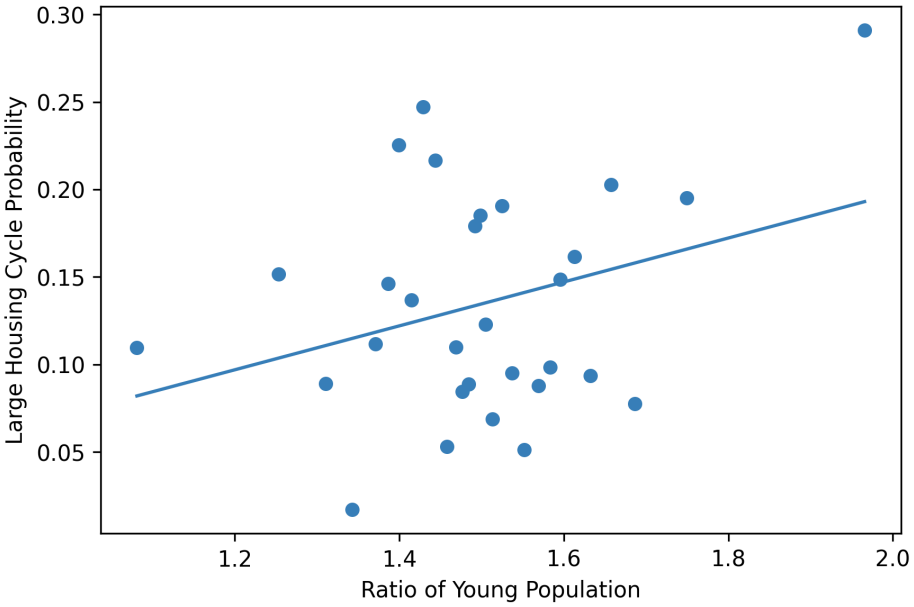
We estimate a sequence of linear probability model (LPM) specifications to examine whether locations with a higher young-to-old population ratio are more vulnerable to large housing booms. In both the cross-country and U.S. county analyses, we begin with parsimonious specifications and then progressively introduce additional controls and fixed effects to assess the robustness of the estimated relationship.

Our baseline specification is given by

$$\mathrm{LHB}_{i,t} = \beta_0 + \beta_1 \mathrm{YO}_{i,t} + Z'_{i,t}\boldsymbol{\phi} + \alpha_i + \gamma_t + \varepsilon_{i,t}, \tag{4}$$

[2]The country-level plot is reported without population weights because each country is treated as a national housing-market regime, with its own macroeconomic, institutional, and mortgage-market environment. The U.S. county-level plot, by contrast, uses county population weights. Counties differ substantially in market size, and treating the smallest and largest counties symmetrically can give disproportionate influence to idiosyncratic demographic variation in small local markets. Because the proposed mechanism operates through household demand and local housing-market participation, population provides a natural proxy for the size of the relevant housing market. Appendix Figure 5 reports the corresponding unweighted county-level bin-scatter plot; the relationship is weaker but remains positive.

(a) Country Panel



(b) U.S. County Panel

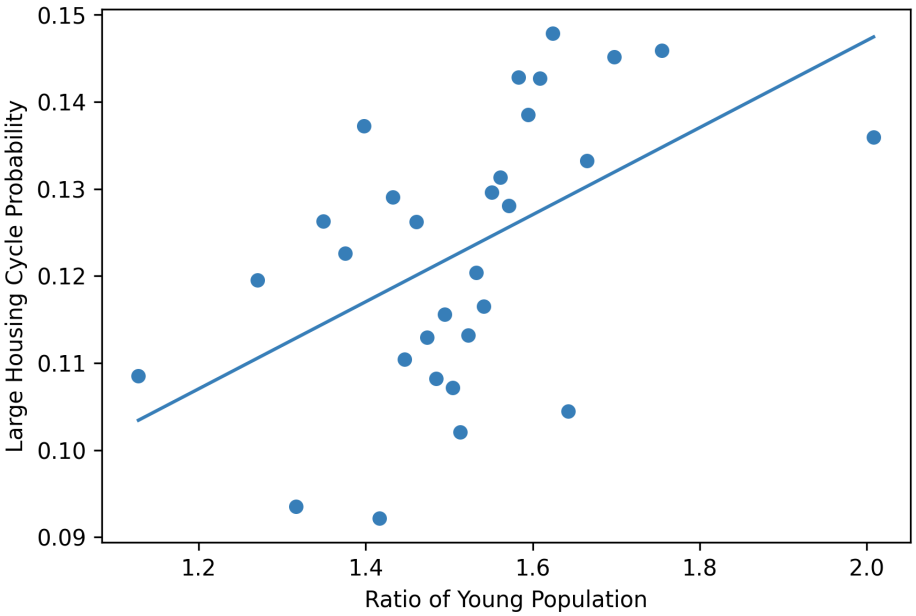


Figure 1: Bin Scatter Plot of Large Housing Booms and Young-to-Old Population Ratio

Notes: This figure presents bin-scatter plots of the large housing boom probability against the young-to-old population ratio. Panel (a) shows the country-level result and Panel (b) shows the U.S. county-level result. Observations are grouped into 30 equal-sized bins based on the young-to-old ratio, and each plot displays the mean large housing boom probability within each bin along with a linear fit. Excluding the single highest-leverage country-year observation (Japan, 1970) leaves the fitted slope in Panel (a) essentially unchanged (0.125 vs. 0.121).

where $\text{LHB}_{i,t}$ is an indicator equal to one if location i experiences a large housing boom at time t, $\text{YO}_{i,t}$[3] is the young-to-old population ratio, $Z_{i,t}$ is a vector of control variables, α_i denotes location-specific fixed effects when included, and γ_t denotes time fixed effects when included. Standard errors are clustered at the location level throughout.

In the cross-country analysis, the control vector includes log real GDP per capita, while in the U.S. county analysis it includes housing supply elasticity. Across specifications, we progressively add controls, location fixed effects, and time fixed effects. Because the LPM is linear, the coefficient β_1 is directly interpretable as the change in the probability of a large housing boom associated with a one-unit increase in the young-to-old ratio, holding other covariates constant. This approach allows us to assess whether the relationship between demographic structure and large housing booms is robust to various specifications without the complications arising from nonlinear models.

Tables 2 and 3 report the results. Across both the cross-country and U.S. county samples, the coefficient on the young-to-old ratio is positive and statistically significant in all specifications, indicating that locations with younger demographic structures have a higher incidence of large upward deviations of house prices from trend.

Table 2 shows a positive, statistically significant relationship in the country panel. The coefficient on the young-to-old ratio is positive in all five specifications, ranging from 0.045 in the most parsimonious specification to 0.231 in the specification with country fixed effects and GDP per capita. In the most demanding specification with both country and time fixed effects (column 5), a one-unit increase in the young-to-old ratio is associated with an 18.6 percentage-point increase in the probability of a large housing boom. Because this variable is a ratio rather than a standardized regressor, the one-unit coefficient is best interpreted as the slope of the boom-probability relationship. The estimated relationship remains statistically significant after introducing country and time fixed effects, suggesting that the result is not driven solely by time-invariant cross-country heterogeneity or common global shocks. GDP per capita enters positively and is statistically

[3]We use the contemporaneous young-to-old ratio because it is a slow-moving demographic state variable. Since the ratio is constructed from broad twenty-year age bins, its one-year lag is mechanically close to the contemporaneous value, especially in the cross-country panel where annual changes in age composition are limited. Hence, we interpret β_1 as a conditional demographic gradient in housing-boom vulnerability, not as the causal effect of an exogenous demographic shock.

significant at the 5% level in both specifications in which it is included, indicating that higher income levels are also associated with a greater likelihood of large housing booms, though the demographic composition variable remains the more robust predictor.

Table 2: Linear Probability Model: Country Panel

	(1)	(2)	(3)	(4)	(5)
YO ratio	0.045*	0.160***	0.231***	0.147**	0.186**
	(0.023)	(0.030)	(0.052)	(0.064)	(0.077)
GDP per Cap.			0.083**		0.323**
			(0.039)		(0.118)
Observations	1,262	1,262	1,225	1,262	1,225
Country FE	No	Yes	Yes	Yes	Yes
Time FE	No	No	No	Yes	Yes
R^2	0.0045	0.0252	0.0298	0.1895	0.2058

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the country level.

Table 3 shows that the same qualitative relationship holds in the U.S. county panel. The coefficient on the young-to-old ratio is positive and highly significant across all specifications, ranging from 0.013 in the time-FE-only specification (column 3) to 0.075 in the pooled specification (column 1). In the specification with both county and time fixed effects (column 5), a one-unit increase in the young-to-old ratio is associated with a 4.3 percentage-point increase in the probability of a large housing boom. Supply elasticity enters with a negative coefficient in column (4), consistent with the view that more elastic housing supply dampens the translation of demand shifts into house price fluctuations, though the estimate is not statistically significant. The substantial rise in R^2 in specifications with time fixed effects suggests that common temporal shocks explain an important share of boom incidence in U.S. counties, although the demographic effect remains statistically meaningful.

The difference in the estimated magnitude of the young-to-old population ratio effect across the country and U.S. county panels may reflect several factors. One possible explanation is that the country panel reflects slow-moving differences in national demographic regimes, housing-finance institutions, and macroeconomic conditions. In contrast, the county panel exploits within-country variation across local housing markets, where population mobility, commuting patterns,

and idiosyncratic shocks can weaken the mapping from age structure to local housing booms. More broadly, the comparison suggests that while the magnitude of the coefficients differs across samples, the positive relationship between the young-to-old ratio and the likelihood of a large housing boom holds across different empirical settings.

Table 3: Linear Probability Model: U.S. County Panel

	(1)	(2)	(3)	(4)	(5)
YO ratio	0.075***	0.044***	0.013***	0.022***	0.043**
	(0.001)	(0.006)	(0.004)	(0.006)	(0.019)
Supply Elasticity				−0.002	
				(0.002)	
Observations	101,567	101,567	101,567	37,205	101,567
County FE	No	Yes	No	No	Yes
Time FE	No	No	Yes	Yes	Yes
R^2	0.1178	0.0159	0.2887	0.3198	0.3046

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All regressions are weighted by county population. Standard errors are clustered at the county level. Column (4) includes housing supply elasticity. Column (5) omits the elasticity because the Saiz (2010a) measure is time-invariant and is absorbed by the county fixed effects.

2.6 Young Applicants and Mortgage-Market Participation During Booms

The previous regression results connect the two ends of the proposed mechanism: locations with younger demographic structures have higher boom incidence. They do not, however, identify the intermediate demand margin through which demographics may translate into larger housing booms.

To examine this margin directly, we turn to mortgage application data from HMDA. HMDA provides loan-level data on mortgage applications and their outcomes for covered financial institutions. The data span a broad segment of the U.S. mortgage market and include information on loan purpose, property location, applicant characteristics, and applicant age in decade-based bins.[4] This makes HMDA a natural source for measuring the extensive-margin demand for housing fi-

[4]Estimates from the Consumer Financial Protection Bureau (CFPB) indicate that HMDA captures about 92 percent of mortgage originations nationally (2017). Estimates from the Federal Reserve Bank of Minneapolis suggest coverage above 90 percent in metropolitan counties and about 80 percent in remote nonmetropolitan counties (2015).

nance: an application reflects a household's attempt to obtain mortgage credit, while originated loans provide a narrower measure of completed housing-finance transactions.

We construct a county–age-group–year panel over the 2018–2024 HMDA sample period, which is the period over which applicant age is reported in the public HMDA files. The outcome is mortgage applications per 1,000 age-specific population in county c, age group a, and year t. We identify housing boom county-years using the same large-boom definition as in the country and county analyses above: the cyclical component of real house prices must exceed one standard deviation above its location-specific trend. We then estimate the following triple-difference specification:

$$Y_{c,a,t} = \beta \left(\text{Boom}_{c,t} \times \text{Young}_{a} \right) + \alpha_{c,a} + \gamma_{t,a} + \delta_{c,t} + \varepsilon_{c,a,t}, \tag{5}$$

where $Y_{c,a,t}$ denotes applications per 1,000 age-specific population. The fixed effects are county-by-age effects $\alpha_{c,a}$, year-by-age effects $\gamma_{t,a}$, and county-by-year effects $\delta_{c,t}$. The specification is highly saturated: the county-by-year fixed effects absorb all county-year shocks and any level effect of the boom indicator itself, including local credit conditions, local house-price growth, and other time-varying county-level factors common across age groups. Identification therefore comes from whether, within the same county-year, young applicants increase mortgage-market activity more than older applicants when that county is in a boom. The coefficient β measures the differential increase in young applicants' application intensity in boom county-years relative to the older comparison group. We interpret this specification as evidence on within-county-year age-group differences in mortgage-market participation, rather than as a fully causal estimate of the effect of boom onset on young households.

Table 4 reports the results. Across columns (1)–(4), the estimated boom-period young-minus-old differential ranges from 1.00 to 1.69 additional mortgage applications per 1,000 age-specific population. The natural denominator for interpreting these coefficients is the comparison group's application intensity in boom county-years, because the regression coefficient is the incremental young-versus-old gap during boom episodes. Relative to the older comparison groups' mean boom-period application intensity, these estimates correspond to roughly 4.3% to 6.6% higher application intensity for young groups. Using the older groups' full-sample mean application

intensity instead yields a somewhat larger range of about 5.3% to 8.0%. These results indicate that young applicants disproportionately increase their mortgage-market activity during housing booms relative to older applicants. The finding complements the LPM evidence by showing not only that younger areas have higher boom incidence, but also that young cohorts are themselves more active participants in the mortgage market precisely when booms occur. This pattern is consistent with the life-cycle interpretation that younger households, who are at the stage of household formation and initial home purchase, respond more strongly to boom-period conditions.

Table 4: HMDA Baseline: Application Intensity in Boom County-Years

	(1)	(2)	(3)	(4)
Young group	25–34	25–44	25–34	25–44
Comparison group	45–74	45–74	55–74	55–74
$\text{Boom}_{c,t} \times \text{Young}_a$	1.002***	1.346***	1.335***	1.686***
	(0.179)	(0.227)	(0.235)	(0.264)
Observations	76,328	95,587	57,161	76,420
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,113	13,892	8,349	11,113
R^2	0.8896	0.8755	0.8991	0.8798

Notes: The outcome is mortgage applications per 1,000 age-specific population. Each column estimates equation (5) using the indicated young and comparison groups. Columns (3)–(4) use an older comparison group as a robustness check. Regressions are weighted by age-specific population. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

One potential concern is that the increase in mortgage applications by younger applicants does not necessarily reflect actual housing demand. For example, younger households may submit more applications per property by shopping across multiple lenders, or their applications may be less likely to translate into completed home purchases. While HMDA records only applications that enter the lender's formal underwriting process, that is, oral or written requests submitted according to the institution's application procedures, rather than informal information inquiries, it is nonetheless useful to examine whether our results are robust to alternative outcomes and comparison groups.

To address this concern, we re-estimate our baseline specification using alternative outcome measures: originated loans and denied applications. Table 5 reports the results. When restricting the outcome to originated loans, the estimated coefficients decline in magnitude but remain large and statistically significant. This stricter outcome indicates that the increased application activity among young applicants during booms translates into higher realized borrowing, rather than reflecting only multiple applications or unsuccessful credit searches. Denied applications also rise significantly for young applicants during boom periods, suggesting that part of the increased activity reflects a broader expansion in credit-seeking behavior. Quantitatively, the origination margin accounts for the bulk of the baseline application response, suggesting that the result largely reflects completed mortgage transactions rather than unsuccessful credit searches. This pattern supports the interpretation that effective housing demand by young households rises disproportionately during boom periods. Additional robustness checks are reported in Appendix C.2.

Another concern is reverse causality: young households might move into counties that are already booming and apply for mortgages there, so that booms attract the young rather than the young amplifying booms. To check this, we estimate an event study of the young-old application gap around each county's first boom onset (Appendix C.1). The gap rises into the boom and remains elevated through it, and re-centering the event study on a clean pre-boom year leaves the onset and post-onset differences intact. This timing fits a demand response that leads the price boom rather than in-migration that follows it. Furthermore, the outcome variable is scaled by age-specific population, so in-migration raises the denominator as well as the numerator. We therefore read reverse causality as unlikely to account for the result.

Beyond young households' extensive-margin response, Appendix C.3 examines whether they also amplify housing booms through an intensive-margin channel by taking on higher leverage during boom periods. The estimated effects are modest in magnitude but statistically significant: relative to older borrowers, young borrowers exhibit higher loan-to-value and debt-to-income ratios in boom county-years.

Table 5: HMDA Robustness: Mortgage Application Outcomes

	(1)	(2)	(3)	(4)
Young group	25–34	25–44	25–34	25–44
Outcome	Originations	Originations	Denials	Denials
Comparison group	45–74	45–74	45–74	45–74
$\text{Boom}_{c,t} \times \text{Young}_a$	0.820***	0.968***	0.071***	0.105***
	(0.123)	(0.148)	(0.021)	(0.024)
Observations	76,328	95,587	76,328	95,587
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,113	13,892	11,113	13,892

Notes: Outcomes are measured per 1,000 age-specific population. Each column estimates equation (5) using the indicated outcome and age-group definition. Regressions are weighted by age-specific population. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

2.7 Monetary Easing and the Age Gradient of House-Price Responses

The cross-sectional results show that younger areas experience larger housing booms. We now ask whether local age structure also shapes how house prices respond to an identified monetary shock. This turns the static association into a dynamic response, and it delivers the moment we later confront the model with.

In particular, we estimate in reduced form the channel through which monetary policy raises housing demand and prices via credit supply. Credit supply shocks have been prominently identified in the house-price literature as a driver of housing booms (Mian and Sufi, 2009; Di Maggio and Kermani, 2017). We use the Large-Scale Asset Purchase (LSAP) easing shock, one of the identified monetary policy surprises of Swanson (2021), as our proxy for the credit-supply shock, since large-scale asset purchases target mortgage-backed securities (MBS) and long-term Treasuries directly, compressing MBS and mortgage spreads on impact.

We estimate the response on a quarterly panel of metropolitan areas. The shock Z_t is a high-frequency identified LSAP easing surprise, signed so that a positive value eases policy. [5] We use the Federal Housing Finance Agency (FHFA) quarterly metropolitan house-price index, deflated to real terms. The outcome at horizon h is the cumulative real house-price change, $100\,[\log P_{m,t+h} - \log P_{m,t-1}]$, for $h = 0, \ldots, 12$ quarters. The panel matches 366 metropolitan areas to a fixed county-to-metro geography and to the 2000 age structure. We also control for local housing-supply elasticity, denoted Supply_m, using the index of Baum-Snow and Han (2024); Appendix D.3 details its construction and a Saiz (2010b) robustness check. The variable YO_m is the standardized young-to-old population ratio, using the same definition as in Section 2.

We run Jorda (2005) local projections of the cumulative house-price response on the shock interacted with local age structure,

$$\log P_{m,t+h} - \log P_{m,t-1} = \alpha_m^h + \delta_t^h + \beta_h \left(Z_t \times \text{YO}_m\right) + \gamma_h \left(Z_t \times \text{Supply}_m\right) + \mathbf{X}'_{m,t}\Lambda_h + u_{m,t}^h, \qquad h = 0, \ldots, 12. \tag{6}$$

[5] We do not instrument the cross-sectional mortgage-rate response with LSAP (the interacted first stage is weak, $F \approx 0.1$, because the fixed-2000 young/old exposure is time-invariant). Instead we report the identified LSAP $\times$ YO reduced form and the national LSAP$\rightarrow$mortgage-spread pass-through separately, and let the calibrated model translate the spread into a house-price age gradient.

The area effects α_m^h and time effects δ_t^h absorb the level of the aggregate shock and the level of age structure, so β_h is identified from differences in the response across areas with different age structures. It measures the age gradient of the h-quarter house-price response to easing, in percentage points per standard deviation of YO_m. We interact the shock with Supply_m so that this gradient reflects age structure rather than differences in supply elasticity. The design identifies the gradient, not the aggregate level of the response.

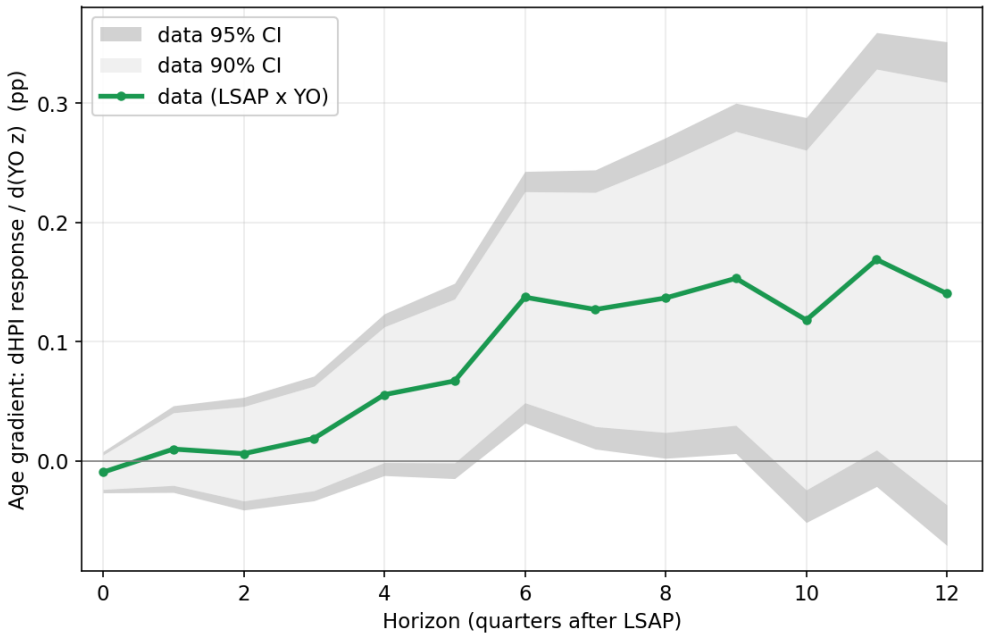


Figure 2: Age Gradient of the House-Price Response to an LSAP Easing

Notes: Local-projection coefficient β_h from (6) by horizon, with 90 and 95 percent confidence bands. The gradient is the change in the h-quarter cumulative house-price response per one standard deviation increase in the young-to-old ratio.

Figure 2 plots $\{\beta_h\}$. The gradient is small per shock but spans a wide cross-section. A one-standard-deviation LSAP easing raises an area's two-year house-price response by 0.14 percentage points for each standard deviation of its young/old ratio. That ratio has a cross-sectional standard deviation of 0.33 (mean 1.65). For instance, an older industrial metro such as Pittsburgh (−1.2 SD) lies more than two standard deviations from younger metros such as Houston (+0.9 SD) or San Diego (+1.1 SD). A single easing surprise therefore opens a two-year house-price gap of roughly 0.3 percentage points between such metros (0.14×2.3). This is modest per shock, but with LSAP

shocks occurring in nearly every quarter from 2008 to 2014, the age differential compounds into an economically meaningful gap.[6]

3 A Housing OLG Model

The empirical section documents that large housing booms are more likely where the young-to-old population ratio is higher. This section asks what economic mechanism can produce that pattern. The model is not used to establish the demographic gradient, which is estimated directly in the data; it is used to interpret it, and to ask whether an age structure alone can make the same shock generate a larger house-price response. Because a boom is recorded when prices rise past a fixed threshold, a larger response to the same shock is precisely what makes booms more likely in younger regions. We therefore build a life-cycle economy with tenure choice in which the age distribution is the only object that varies across the experiments of interest.

3.1 Overview

The economy is populated by overlapping generations of households who live through a deterministic life cycle and choose each period between owning and renting a house. Households consume a non-housing good and housing services, save in a risk-free liquid asset, and finance a home purchase with a long-term fixed-rate mortgage subject to a loan-to-value limit at origination. The stock of owner-occupied housing is fixed, so the house price clears the owner-occupied market each period; rental housing is supplied elastically by an outside sector at a rent proportional to the price. Households face idiosyncratic income risk, but the economy has no aggregate uncertainty: the age structure, entered through the population weights on each cohort, is the only object we vary to represent younger and older societies.

Two features do the work. First, tenure is a discrete owner-renter choice, so a cohort's contribution to housing demand is governed by the mass of households near the ownership margin

[6]The quarterly LSAP shock is nonzero in every quarter of the 2001Q1–2019Q2 regression sample (SD = 1.07), and several quarters are far larger than a typical shock: the QE1-announcement quarter (2009Q1, which includes the March 18, 2009 balance-sheet expansion) is 4.9 SD, the Operation Twist quarter (2011Q3) is 1.4 SD, and the taper-surprise quarter (2013Q3, which includes the September 18, 2013 decision to postpone tapering) is 2.6 SD.

rather than by a smooth demand curve. Second, the young are concentrated near that margin, because they are the cohorts deciding whether to buy their first home. We present the environment, the household problem, and the equilibrium, and then calibrate the model; the two channels through which age structure shapes housing demand are derived analytically in Appendix E. The quantitative exercise, the house-price response to a credit-supply shock, is deferred to Section 3.6.

3.2 Environment

Demographics. Time is discrete and a period is one year. Households live through cohorts $g = 1, \ldots, G$ and die with certainty after cohort G; a cohort-g household faces age-specific survival rates taken from standard U.S. life tables. All cohorts are alive at each date. The population weight of cohort g is π_g, with $\sum_g \pi_g = 1$, and we generate a society's age structure from a single cohort-growth rate n,

$$\pi_g \;\propto\; (1+n)^{-(g-1)}, \tag{7}$$

so that $n > 0$ describes a young society and $n < 0$ an old one. This one-parameter family is the comparative static of the paper: the young-to-old ratio in the data maps into n, and every experiment holds preferences, input prices, and the shock fixed while moving n. Long-run demographic variation of this kind has been linked to housing demand since Mankiw and Weil (1989); our contribution is to trace it through the tenure margin.

Preferences. A household with m periods of remaining life has expected utility

$$U \;=\; \mathbb{E}\sum_{k=1}^{m} \beta^{\,k-1}\big[\,\ln c_k \;+\; \omega\, s_k\,\big] \;+\; \beta^{\,m}\,\phi_B\,\ln\!\big(W_{\text{beq}}\big), \tag{8}$$

where c_k is non-housing consumption, $s_k \in \{0,1\}$ indicates home ownership, $\omega > 0$ is the per-period utility premium from owning, and $\phi_B \geq 0$ is a warm-glow motive over terminal wealth W_{beq} (including the value of a house left at death). The ownership premium ω is a reduced-form stand-in for the non-pecuniary and tax advantages of owning, and we discipline it below by the aggregate home-ownership rate.

Endowments and income risk. During working life a household's income is

$$y_{i,g,t} \;=\; z_i \, e_g \, \varepsilon_{i,t}, \tag{9}$$

the product of a permanent type z_i drawn once at birth, a deterministic age profile e_g, and an independent transitory shock $\varepsilon_{i,t}$.[7] The age profile is hump-shaped and taken from the 2019 Survey of Consumer Finances (SCF), with retirement income a fixed fraction of pre-retirement income. Idiosyncratic income risk is the only uncertainty; there are no aggregate shocks.

Housing and rental markets. Housing is indivisible: a household either owns one unit ($s = 1$) or rents ($s = 0$). The stock of owner-occupied housing is fixed at $\bar{H}$ per capita, so its price p_t is determined by market clearing (Section 3.4). Rental units are supplied elastically by an outside sector, separate from the owner stock, at a rent proportional to the price, $\text{rent}_t = \chi \, p_t$. Owners pay a per-period maintenance cost $\mu \, p_t$, and buying and selling incur proportional transaction costs κ_b and κ_s. The indivisibility and the transaction costs are what place the young at the ownership margin and give the extensive margin its role (Ortalo-Magne and Rady, 2006; Chambers et al., 2009).

Mortgages. A purchase is financed by a mortgage originated at the time of purchase, subject to the loan-to-value limit $b \leq \lambda \, p_t$. The mortgage carries a fixed rate set at origination and fully amortizes over the remaining life, so the balance reaches zero at death and a household that sells early repays the outstanding balance. Because the amortization horizon shortens with age, the annual payment rises steeply for older buyers, and the last cohort, which has no remaining periods to amortize, can buy only with cash. There is no default, no refinancing, and no home-equity extraction; the mortgage is the sole borrowing instrument, and liquid savings $a \geq 0$ cannot be negative.

[7] The permanent type is a fixed effect: it carries no innovations over the life cycle, so income risk does not accumulate as it would under a persistent process (Guvenen et al., 2015). This keeps z a household's fixed lifetime rank relative to the ownership margin. Both z and ε are lognormal, with dispersions set in Section 3.5.

3.3 Household problem

A household enters a period with liquid assets a, previous tenure s_{-1}, an outstanding mortgage balance B, and permanent type z, and draws the income shock ε. It chooses consumption c, tenure $s \in \{0, 1\}$, new borrowing b if it buys, and next-period assets $a' \geq 0$, subject to the budget constraint

$$c \;+\; \frac{a'}{R} \;+\; x^{h}(s_{-1}, s, B) \;=\; y_{g} \;+\; a, \qquad c > 0,\; b \leq \lambda p, \tag{10}$$

where R is the gross return on the liquid asset and x^{h} is the net housing outlay, which equals the rent χp for a continuing renter, maintenance μp plus the mortgage payment for a continuing owner, the down payment plus transaction cost $p(1 + \kappa_b) - b$ for a buyer, and the sale proceeds net of transaction cost and outstanding balance for a seller. Letting V_g denote the value of a cohort-g household, the problem is

$$V_{g}(a, s_{-1}, B, z) = \max_{c, s, b, a'} \left\{ \ln c + \omega\, s + \beta\, \rho_{g}\, \mathbb{E}_{\varepsilon'}\, V_{g+1}(a', s, B', z) \right\}, \tag{11}$$

subject to (10) and the amortization rule for B', where ρ_g is the survival rate. The terminal condition applies the warm-glow bequest to the final period's wealth. The problem is solved by backward induction on a grid for (a, B, z); we report the numerical scheme in the appendix.

3.4 Equilibrium

Given the age structure $\{\pi_g\}$, the mortgage rate, and the type distribution, a stationary equilibrium is a price p^{*} and household policies such that households optimize and the owner-occupied market clears,

$$\sum_{g=1}^{G} \pi_{g} \int \mathbf{1}\{s_{g}^{*}(z) = 1\}\, dF(z) \;=\; \bar{H}. \tag{12}$$

Because ownership demand is decreasing in the price, the equilibrium price is unique. Comparing stationary equilibria across values of n isolates the effect of the age structure on the price level; comparing responses to a common shock across values of n, the exercise of Section 3.6, isolates its effect on the price response.

3.5 Parameterization

Following Berger et al. (2018), we split parameters into those set externally and those calibrated internally, and report both in Table 6. Externally, the real return, mortgage spread, and loan-to-value limit take standard values; the rent-to-price ratio follows the aggregate rent-price evidence in Davis et al. (2008) and Piazzesi and Schneider (2016); the maintenance-and-property-tax rate follows the user-cost decomposition in Poterba (1991), combining economic depreciation (Harding et al., 2007) with an effective property tax; and the transaction costs follow life-cycle housing models such as Kaplan et al. (2020). The age-earnings profile and the owner-occupied stock come from the 2019 SCF, and survival rates from the U.S. life table. We set the transitory income dispersion externally and back out the permanent dispersion so that within-age log-income dispersion matches the SCF.[8] Internally, the discount factor, ownership premium, and bequest strength are chosen by weighted minimum distance to 24 life-cycle moments from the 2019 SCF, the home-ownership rate, liquid assets, home equity, and owner loan-to-value, each in six age groups.

Figure 3 shows the fit of the calibrated model against the 2019 SCF, by age bin: homeownership (panel (a)), liquid assets relative to income (panel (b)), home equity relative to income (panel (c)), and the owner loan-to-value ratio (panel (d)). Circles are model values and squares are SCF data. With only three free parameters (β, ω, ϕ_B) disciplining the 24 targeted moments, the model tracks the homeownership (a) and loan-to-value (d) profiles closely across ages.

The two clearest discrepancies are in the wealth moments, and both trace to the same source: the model is calibrated in its stationary equilibrium, where the house price is constant, so home equity accumulates only through mortgage amortization and never through price appreciation. Older SCF households, by contrast, hold decades of realized house-price gains on top of amortization, so the model's shortfall in home equity (c) widens steadily with age. With that channel closed off, the calibration's only remaining margin for matching older households' overall wealth position is the liquid asset, which is why the model overshoots liquid assets most at the oldest ages (b).

[8] The transitory dispersion exceeds the pure-transitory consensus from the Panel Study of Income Dynamics (PSID; sd $\approx$ 0.20–0.25; Kaplan and Violante, 2014; Floden and Lindé, 2001) because the process omits a persistent component that the iid shock must absorb instead. The permanent/transitory split is therefore not separately identified; results should be read as robust to, not pinned down by, this split.

Table 6: Parameterization

	Parameter	Value	Source / target
Panel A. Set externally or to external evidence			
r	real return	2%	standard
κ_m	mortgage spread	2.8%	FRED[a]
λ	loan-to-value limit	0.80	regulatory
χ	rent-to-price ratio	5%	price-rent evidence
μ	maint. + property tax	2.5%	housing user cost
κ_b	purchase cost	2%	life-cycle housing lit.
κ_s	sale cost	5%	life-cycle housing lit.[c]
σ_ε	transitory income s.d.	0.35	externally set[b]
σ_z	permanent income s.d.	0.85	estimated from SCF
e_g	age-earnings profile	—	SCF
$\bar{H}$	owner-occupied stock	0.669	SCF home-ownership rate
	survival rates	—	2000 U.S. life table
Panel B. Calibrated to match SCF life-cycle moments			
β	discount factor	0.888	
ω	ownership premium	0.114	home-ownership, liquid assets,
ϕ_B	bequest strength	5.12	home equity, LTV by age (24 moments)

Notes: Panel B parameters are chosen by weighted minimum distance to 24 moments from the 2019 SCF (home-ownership rate, liquid-assets-to-income, home-equity-to-income, and owner loan-to-value, each in six age groups). [a] 30-year FRM - FFR, 1971–2024 average. [b] σ_ε is externally set rather than estimated; σ_z is then backed out so that the model's within-age standard deviation of log income equals the SCF value of 0.92. [c] The sale cost κ_s = 5% is below the customary 6% realtor commission; we interpret it as net of some fees and treat results as provisional in this parameter.

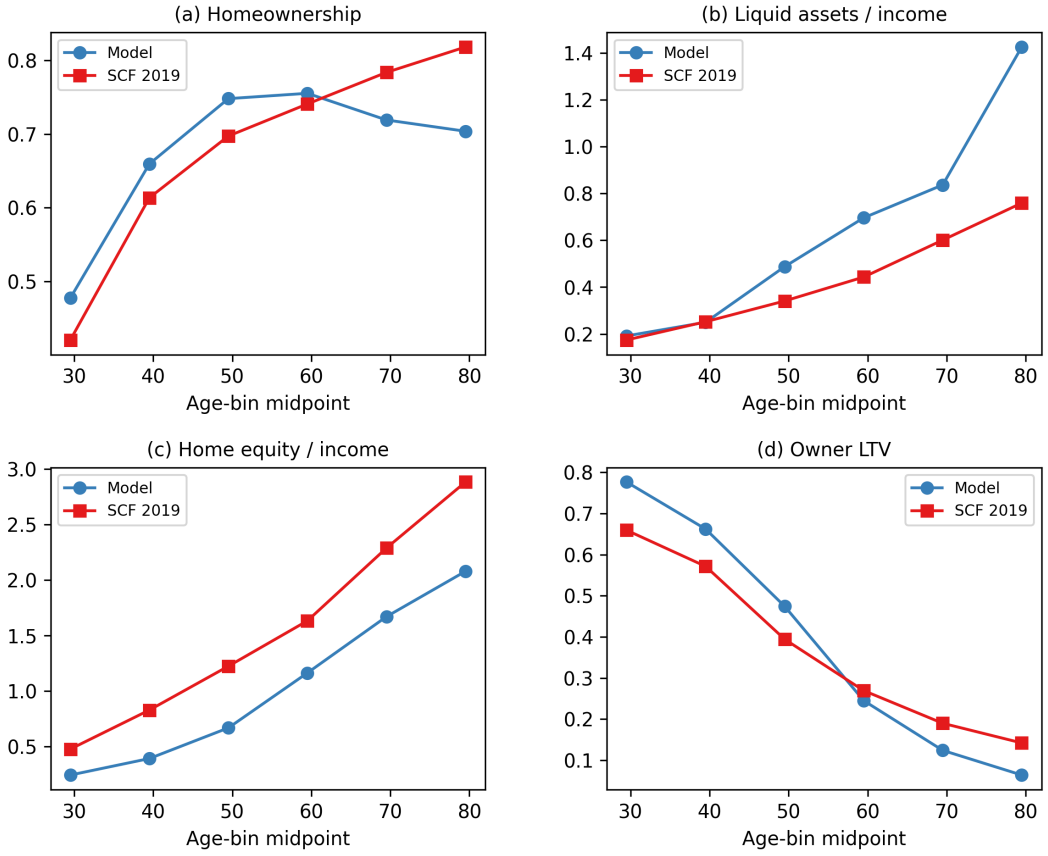


Figure 3: Life-Cycle Calibration: Model versus SCF

Notes: The figure reports the fit of the calibrated model (circles) against the 2019 SCF (squares), by age-bin midpoint, for (a) the homeownership rate, (b) liquid assets relative to income, (c) home equity relative to income, and (d) the owner loan-to-value ratio. These are the 24 moments targeted by weighted minimum distance in Table 6, Panel B.

3.6 Quantitative Exercise: Age Structure and the House-Price Response to a Credit Shock

We now turn to the model experiment to see whether age structure alone amplifies the house-price response. As noted in Section 2.7, we focus on the credit shock long discussed in the literature as a primary driver of house-price booms. The model has no primitive that directly implements a credit shock, so we proxy it with a compression in the mortgage spread κ_m (the mortgage rate net of the risk-free rate).

Population mapping. We vary the model's population composition across young and old societies by moving a single parameter, the annual cohort growth rate n, holding preferences, technology, the mortgage contract, and the equilibrium price map fixed.[9] The model young/old ratio (ages 25–34 relative to 55–64) is therefore 1.64, 1.15, and 0.80 across the three economies, matching the construction of the empirical ratio. To place the model on the same scale as the reduced form, we standardize this ratio by the population-weighted cross-sectional mean (1.65) and standard deviation (0.33) of the U.S. MSA young/old distribution; the young and old economies differ by about 2.5 standard deviations of the young/old ratio, and every model–data comparison below is expressed per one standard deviation.

Mapping the shock. We take the estimated pass-through from the LSAP surprise to the conventional mortgage spread and feed its path into κ_m.[10] A one-standard-deviation easing compresses the spread by about 10 basis points after one year, 6 basis points after two years, and returns it to baseline by the third year. Existing owners keep the coupon locked at origination; only new buyers face the lower offer rate; and we roll the age-by-tenure distribution forward each year. The shock is unexpected and one-time, and we re-clear the house price each period, so the economy leaves and

[9] The two economies differ only in the stationary age distribution. Cohort weights satisfy $\pi_g = \pi_{g-1}\,\rho_{g-1}/(1+n)$, where ρ_g is the survival rate defined above, and we set $n = +1.2\%$ ("young"), 0% ("baseline"), and -1.2% ("old") per year. The implied population shares of ages 25–34, 55–64, and 75–84 are (0.26, 0.16, 0.06) in the young economy and (0.15, 0.19, 0.13) in the old economy.

[10] We adjust only the spread, not the model's risk-free rate, for two reasons. First, LSAP easing works through mortgage and mortgage-backed-security spreads rather than the policy rate. Second, a spread shock isolates the channel of interest: the mortgage rate is $r + \kappa_m$, so cutting r would also move the return on liquid saving, mixing a consumption-saving channel into the purchase decision.

returns to the stationary equilibrium of Section 3.4. We then read the model's house-price response for young, baseline, and old societies and take its slope against the standardized young-to-old ratio used in the data. This places the model and the data on the same ruler. The mapping gives the empirical LSAP response a purchase-cost mechanism.[11]

Baseline response. Figure 4 shows the baseline gradient, where households expect no price change ($\hat{b}_t = 0$), against the empirical estimate. The gradient rises to about 0.055 percentage points per standard deviation after one year, close to the data value of 0.056, so younger societies respond more on impact: the peak house-price response is about 1.34% in the young society against 1.20% in the old. The response then fades as the spread reverts, and the gradient falls to 0.025 and 0.019 by years two and three. The data gradient instead keeps rising, to about 0.14. The model reproduces the impact age gradient but not its persistence, and this gap is its main shortfall.

Adaptive expectations. To generate persistence in the house-price response to the shock, we introduce a new channel: adaptive expectations on house-price growth.

This device has both empirical and theoretical support: expectations of house-price growth extrapolate from recent price changes (Greenwood and Shleifer, 2014; Glaeser and Nathanson, 2017), and extrapolation of this kind generates the price momentum and boom-bust dynamics that rational models miss (Glaeser and Nathanson, 2017; Barberis et al., 2018). Feeding recent gains back into demand is therefore the natural channel for turning a transitory credit shock into a persistent house-price response, as in the belief-driven credit cycles of Bordalo et al. (2018).

The belief $\hat{b}_t$ updates by a constant gain K on realized growth,

$$\hat{b}_t = (1 - K)\,\hat{b}_{t-1} + K\left(\log p_t - \log p_{t-1}\right). \tag{13}$$

Expected appreciation lowers the user cost of owning, which shifts the ownership cost by an amount proportional to the belief $\hat{b}_{t-1}$. Households' tenure decisions in this model reflect two

[11] It is not a direct instrumental-variables estimate of the spread elasticity of house prices.

channels, capital gains and rent hedging (Appendix E), and adaptive expectations amplify the capital-gain channel, raising the propensity of marginal buyers to become owners.

The dotted line in Figure 4 is the adaptive path; the gradient now rises to 0.094 and 0.120 percentage points by years two and three, tracking the data's rise to 0.137 and 0.140 and staying inside its 90% band, while the impact value holds at 0.055. Learning amplifies the peak response by about a third and extends the boom into a gradual reversal. The model reproduces the untargeted dynamic gradient, including the persistence that the baseline version misses.

Discussion. Overall, the model shows that societies with more young households experience a larger price increase than old societies under a fixed housing supply, because the long-run utility gain from ownership and the rent-hedging channel move more households into ownership in response to the credit shock. That increase, however, fades quickly and does not match the persistence in the data. Once adaptive expectations give households a trend belief in future price appreciation, the model closely reproduces the data's age gradient. This is consistent with Kaplan et al. (2020)'s argument that credit shocks and house-price expectations are not individually sufficient and generate a housing boom only jointly.

This result comes with two caveats. Extrapolative house-price expectations are well documented in survey and experimental evidence, whereas direct evidence for the specific adaptive updating rule used here is more limited. We therefore view adaptive expectations as a tractable first-pass approximation to this broader class of backward-looking, extrapolative beliefs. The model is also simplified along several dimensions relative to frontier OLG housing models (e.g., Kaplan et al., 2020; Berger et al., 2018). Our model assumes a fixed housing supply, no default or foreclosure, a binary tenure choice, and no price dynamics beyond the mapped spread, so it shuts down many channels present in the data. Extending the model along these dimensions is a natural next step.

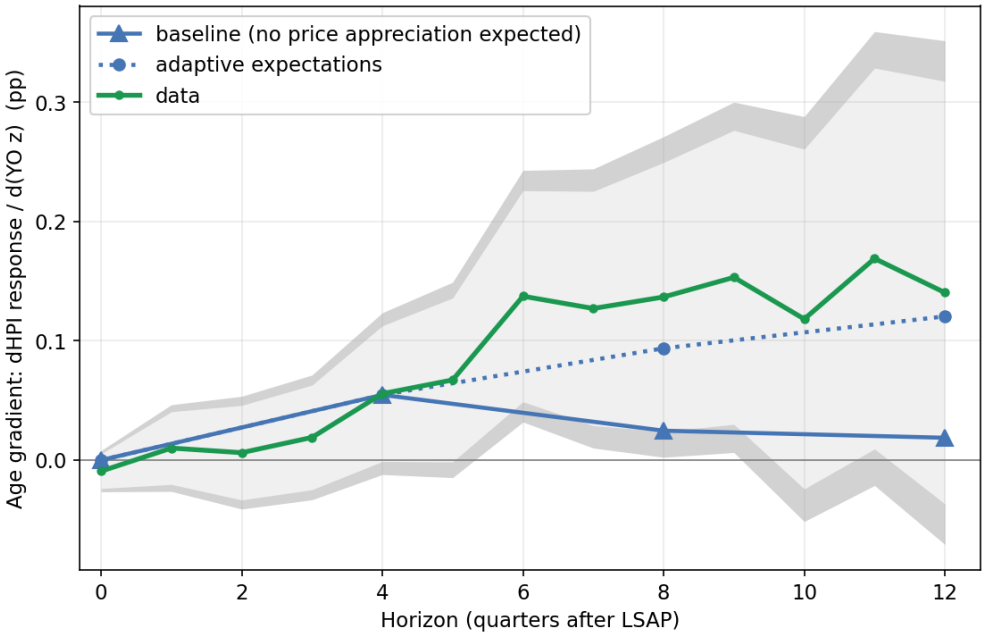


Figure 4: Age Gradient of the House-Price Response: Model versus Data

Notes: By horizon h, the age gradient is the change in the h-period cumulative house-price response to the shock per one-standard-deviation increase in the young-to-old ratio. *Data*: β_h from (6), with 90 and 95 percent confidence bands. *Baseline* and *Adaptive*: the analogous slope, across young, baseline, and old societies, of the model's price response to the mapped mortgage-spread shock, computed with no expected appreciation ($\hat{b}_t = 0$) and with the adaptive belief of (13), respectively.

4 Conclusion

This paper asks whether demographic age structure shapes housing-boom vulnerability. In an OECD country panel and a U.S. county panel, regions with a higher young-to-old population ratio are more likely to experience large upward deviations of real house prices from trend. The association is robust to location and time fixed effects, income and housing-supply-elasticity controls, and alternative boom definitions: a one-unit increase in the young-to-old ratio raises the boom probability by 18.6 percentage points across countries and 4.3 points across U.S. counties. Two further results tie this pattern to mortgage demand and to the response to credit shocks. In mortgage application data, young applicants increase their application intensity 4.3 to 6.6 percent more than older applicants during local booms. And after an identified LSAP easing, house prices rise more, and more persistently, in younger areas. This age gradient builds to about 0.14 percentage points per standard deviation over two to three years.

We interpret these facts through an annual overlapping-generations model with tenure choice. We feed the model a credit shock, proxied by a compression in the mortgage spread, and compare the house-price response across young and old societies. The central model result is that age structure alone amplifies the response: younger societies hold more households at the owner-renter margin, so the same shock moves more of them into ownership and raises the price by more. This impact gradient matches the data. The model does not, on its own, reproduce the persistence of the empirical response. A simple adaptive-expectations extension does, which we offer as suggestive evidence that an expectations channel may be needed to reproduce the full dynamics of the young-society boom. This reading is consistent with Kaplan et al. (2020), where credit and expectations generate a sustained boom only together.

The message is that demographic composition is a demand-side state variable for housing-boom risk. Young populations create larger booms because age composition interacts with life-cycle income and the tenure-adjustment margin, while extrapolative expectations make those booms more persistent. Demographic structure therefore belongs in assessments of housing-market risk and in the analysis of how credit and monetary shocks transmit to house prices.

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A Additional Descriptive Evidence

Figure 5 reports the unweighted version of the U.S. county-level bin-scatter plot. The relationship is weaker than in the population-weighted plot reported in the main text, but it remains positive.

(a) U.S. County Panel: Unweighted

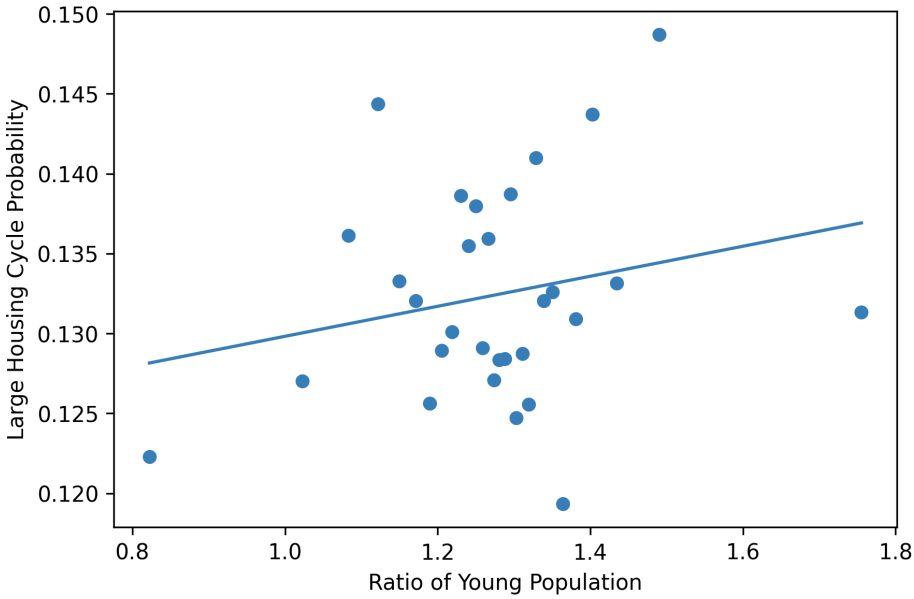


Figure 5: Unweighted Bin Scatter Plot of Large Housing Booms and Young-to-Old Population Ratio: U.S. County Panel

Notes: This figure presents the unweighted U.S. county-level bin-scatter plot of the large housing boom probability against the young-to-old population ratio. Observations are grouped into 30 equal-sized bins based on the young-to-old ratio, and the plot displays the mean large housing boom probability within each bin along with a linear fit.

B Supplementary Logit Specifications

Table 7: Logit Model: Country Panel

	(1)	(2)	(3)	(4)	(5)
YO ratio	0.35**	1.64***	2.27***	1.90***	2.05**
(Marginal Effect)	0.04**	0.19***	0.26***	0.22***	0.24**
GDP per Cap.			0.80**		3.60**
(Marginal Effect)			0.09**		0.43**
Constant	-2.40***	-4.51***	-13.81***	-3.26***	-42.36**
Observations	1,262	1,262	1,225	1,262	1,225
Country FE	No	Yes	Yes	Yes	Yes
Time FE	No	No	No	Yes	Yes
Pseudo R^2	0.0053	0.0352	0.0408	0.1743	0.1884

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the country level.

Table 8: Logit Model: U.S. County Panel

	(1)	(2)	(3)	(4)
YO ratio	0.21***	0.29***	0.15***	0.27***
	(0.03)	(0.04)	(0.05)	(0.07)
(Marginal Effect)	0.02***	0.03***	0.01***	0.02***
Observations	101,567	37,205	101,567	37,205
Supply Elasticity	No	Yes	No	Yes
County FE	No	No	No	No
Time FE	No	No	Yes	Yes
Pseudo R^2	0.0020	0.0034	0.3223	0.3557

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All regressions are weighted by county population. Standard errors are clustered at the county level. County fixed-effect logit specifications were omitted because they are high-dimensional nonlinear FE models prone to perfect prediction.

C HMDA Additional Evidence

C.1 Pretrend

Because the HMDA analysis is interpreted as a triple-difference design, it is useful to examine whether the relative application intensity of young applicants was already moving before a county enters a boom episode. We therefore estimate a short event-study specification around the *first* boom onset in each county. Let $\tau_{c,t}$ denote event time relative to the first year in which the county-level boom indicator switches from zero to one. We estimate

$$y_{c,a,t} = \sum_{k \in \{-2,0,1\}} \beta_k \left(\mathbf{1}\{\tau_{c,t} = k\} \times Young_a \right) + \alpha_{c,a} + \lambda_{t,a} + \mu_{c,t} + \varepsilon_{c,a,t}, \tag{14}$$

where the omitted category is $\tau_{c,t} = -1$. Thus each coefficient measures the young-versus-old application gap at event time k relative to the year immediately preceding boom onset.

This exercise matters because a clean event-study pattern would typically display little movement in the pre-period. If the pre-period coefficients differ materially from zero, the onset-year estimate may partly reflect dynamics that begin before the boom indicator turns on.

The onset and post-onset coefficients are positive and statistically significant, which is consistent with stronger mortgage application intensity among young applicants during boom county-years. At the same time, the coefficient at event time -2 is negative. This means that the young-versus-old application gap was smaller two years before onset than in the year immediately before onset. The event-study evidence therefore suggests some pre-boom dynamics in relative young application intensity rather than perfectly flat pretrends.

There are two reasons to interpret this pattern cautiously. First, the HMDA sample with publicly available applicant age is short (2018–2024), so the number of clean pre-period observations is limited. Second, timing is measured at the annual frequency and the boom indicator is constructed from annual deviations of county house prices from trend. If boom conditions begin to form near the end of year $t-1$, then the immediate pre-period is not a clean untreated benchmark. We therefore interpret the HMDA results as mechanism-consistent evidence on age-specific participation during boom periods, rather than as a clean event-study design with flat pretrends.

Table 9: HMDA Event Study Around First Boom Onset

	(1)	(2)
Young group	25–34	25–44
Comparison group	45–74	45–74
Reference period	$\tau_{c,t} = -1$	$\tau_{c,t} = -1$
$\mathbf{1}\{\tau_{c,t} = -2\} \times Young_a$	-1.231***	-1.114***
	(0.296)	(0.258)
$\mathbf{1}\{\tau_{c,t} = 0\} \times Young_a$	0.715***	1.042***
	(0.197)	(0.223)
$\mathbf{1}\{\tau_{c,t} = 1\} \times Young_a$	1.023***	1.038***
	(0.240)	(0.233)
Observations	87,464	109,796
Counties	3,230	3,230
County-year cells	22,426	22,463
R^2	0.8913	0.8773

Notes: The outcome is mortgage applications per 1,000 age-specific population. Each column estimates equation (14). The omitted event time is the year immediately preceding the first boom onset in a county ($\tau_{c,t} = -1$). Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

C.2 Alternative Boom Definitions and Sample Restrictions

We next assess whether the HMDA results are sensitive to how boom county-years are defined and to simple sample restrictions. The first set of checks tightens the boom criterion by requiring larger deviations of county house prices from trend. The second set trims the estimation sample by dropping county-age cells with very small age-specific populations.

Table 10: HMDA Robustness: Alternative Boom Thresholds

	(1)	(2)	(3)	(4)
Young group	25–34	25–44	25–34	25–44
Comparison group	45–74	45–74	45–74	45–74
Boom definition	> 1.5 SD	> 1.5 SD	> 2.0 SD	> 2.0 SD
$\text{Boom}_{c,t} \times \text{Young}_a$	0.950***	1.223***	1.650***	2.404***
	(0.217)	(0.262)	(0.435)	(0.546)
Observations	76,328	95,587	76,328	95,587
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,113	13,892	11,113	13,892
R^2	0.8895	0.8754	0.8895	0.8754

Notes: The outcome is mortgage applications per 1,000 age-specific population. Each column estimates equation (5) using a stricter boom indicator based on the county-specific standard deviation of detrended annual house prices. Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

These additional checks show that the HMDA result is not driven by a particular boom cutoff or by thin county-age cells. The coefficients remain positive and economically meaningful under stricter boom definitions and when the sample is restricted to larger age groups. Overall, the application-margin result appears robust to alternative boom definitions and basic sample restrictions.

C.3 Risk-Taking Outcomes Among Young Borrowers

The preceding HMDA results show that young applicants increase mortgage-market participation more intensively during boom county-years. We next ask whether this stronger participation is accompanied by riskier mortgage choices among young borrowers. To examine this intensive-

Table 11: HMDA Robustness: Sample Restrictions

	(1)	(2)
Young group	25–34	25–44
Comparison group	45–74	45–74
Restriction	Pop. age > 1,000	Pop. age > 1,000
$\text{Boom}_{c,t} \times \text{Young}_a$	1.014***	1.365***
	(0.181)	(0.231)
Observations	65,814	82,942
Counties	2,697	2,697
County-year cells	18,489	18,494
County-age cells	9,672	12,183
R^2	0.8903	0.8759

Notes: The first two columns require each county-age cell to contain at least 1,000 people. Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. * and *** indicate significance at the 10% and 1% levels, respectively.

margin channel, we re-estimate the HMDA triple-difference specification using age-group averages of loan-to-value (LTV) and debt-to-income (DTI) ratios as outcomes. Specifically, we estimate equation (5) with $y_{c,a,t}$ equal to the county-age-year mean LTV ratio or the county-age-year mean DTI ratio among mortgage applications. As in the baseline HMDA specification, regressions absorb county-age, year-age, and county-year fixed effects, are weighted by age-specific population, and cluster standard errors at the county level.

Because mean LTV and mean DTI could be sensitive to a small number of extreme observations, we focus on a trimmed robustness exercise. For each continuous outcome, we remove county-age-year cells in the top and bottom 1% of the full-sample outcome distribution and then re-estimate the same triple-difference specification. This check is designed to ask whether the risk-taking pattern is driven by outliers or instead reflects a more systematic shift in the composition of young borrowers during booms.

Table 12 shows that the qualitative pattern is stable after trimming extreme LTV and DTI observations. For mean LTV, the coefficient ranges from 0.376 to 0.461, implying that in boom county-years the average LTV of young applicants is about 0.38–0.46 percentage points higher than that of the comparison group, even after removing the top and bottom 1% of county-age-year mean

Table 12: HMDA Risk-Taking During Boom County-Years: Trimmed Mean Outcomes

	(1)	(2)	(3)	(4)
Panel A: Young group = 25–34				
Comparison group	55–74	55–74	45–74	45–74
Outcome	Mean LTV	Mean DTI	Mean LTV	Mean DTI
Trim rule	1% tails removed	1% tails removed	1% tails removed	1% tails removed
$\text{Boom}_{c,t} \times \text{Young}_a$	0.461***	0.115**	0.408***	0.098**
	(0.095)	(0.052)	(0.078)	(0.041)
Observations	56,186	56,590	75,172	75,649
Counties	2,779	2,779	2,779	2,779
County-year cells	19,097	19,098	19,099	19,098
County-age cells	8,333	8,332	11,110	11,110
R^2	0.9374	0.8913	0.9256	0.8727
Panel B: Young group = 25–44				
Comparison group	55–74	55–74	45–74	45–74
Outcome	Mean LTV	Mean DTI	Mean LTV	Mean DTI
Trim rule	1% tails removed	1% tails removed	1% tails removed	1% tails removed
$\text{Boom}_{c,t} \times \text{Young}_a$	0.431***	0.090*	0.376***	0.072**
	(0.100)	(0.047)	(0.082)	(0.035)
Observations	75,222	75,794	94,208	94,853
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,112	11,110	13,889	13,888
R^2	0.9203	0.8814	0.9114	0.8685

Notes: The outcomes are county-age-year mean loan-to-value ratios and county-age-year mean debt-to-income ratios among mortgage applications. Each column estimates equation (5) after removing the top and bottom 1% of the corresponding outcome distribution in the full sample. Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

outcomes. Relative to the comparison groups' weighted mean LTV levels in the estimation sample, this corresponds to an increase of roughly 0.5%. For mean DTI, the coefficients range from 0.072 to 0.115 percentage points, which is economically modest but generally positive and statistically significant in three of the four specifications. Relative to the comparison groups' weighted mean DTI levels, these effects are on the order of about 0.2%–0.3%.

Taken together, these results suggest that the main HMDA finding is not solely an application-margin phenomenon. Young applicants are more active in boom county-years, and conditional on applying, they also appear to tilt modestly toward riskier mortgage positions, especially along the LTV margin.

D Identification of the LSAP Age Gradient

This appendix constructs the monetary shock and estimates the age gradient reported in Section 2.7. It supports the empirical target that Section 3 confronts the model with. The reported object is the reduced-form response of local house prices to an identified easing surprise, interacted with local age structure and with housing-supply elasticity.

D.1 The high-frequency LSAP surprise

We measure large-scale asset purchases with the LSAP factor of Swanson (2021), identified from asset-price movements in narrow windows around scheduled monetary policy announcements. The factor is orthogonal to the current-rate and forward-guidance surprises from the same decomposition, so it isolates variation in asset purchases from conventional and communication policy. We sum the announcement-level surprises to the quarter and sign the series so that a positive value eases policy. The sample runs from 2000 to 2019.[12]

[12]The reduced form is our reported object. We also instrument the change in the 30-year mortgage rate with the surprises and their interactions, to scale the response by realized rate pass-through. The first stage for the interacted mortgage-rate term is weak, and the implied age gradient is negative and insignificant. We therefore report the reduced form and do not interpret an instrumental-variables elasticity.

D.2 Specification

We estimate the local projections in (6) area by area over horizons. Local age structure is the standardized young-to-old population ratio, and we standardize it with the sample population weights. The baseline includes MSA and time effects; a robustness specification replaces the time effects with state-by-quarter effects. We weight by lagged metropolitan population. We cluster standard errors two-way by MSA and quarter, and apply an eigenvalue correction when the finite-sample covariance is not positive definite.

D.3 Housing-supply control

Areas with inelastic housing supply raise prices more for any demand shift, and supply elasticity correlates with age structure. We therefore interact every national shock with a fixed local supply elasticity. The level of the elasticity is absorbed by the metropolitan effect, so the added term is $Z_t \times \text{Supply}_m$. Our baseline elasticity is the housing-unit measure of Baum-Snow and Han (2024), aggregated from census tracts to metro areas under their common-neighborhood-demand rule,

$$\text{Supply}_m = \frac{\sum_i H_i \, \gamma_i/(1+\gamma_i)}{\sum_i H_i/(1+\gamma_i)}, \tag{15}$$

where γ_i is the tract elasticity and H_i the tract housing stock.[13]

The supply control leaves the point estimate almost unchanged but widens its confidence interval. At four quarters the age gradient is 0.056 (s.e. 0.035) with the baseline control and 0.086 (s.e. 0.073) with the Saiz control, both insignificant. The gradient becomes significant at medium horizons: 0.137 ($p = 0.011$) at six quarters and 0.153 at nine quarters. We report the specification with the supply control throughout.

D.4 Remaining threats

Three limits bound the claim. First, among the three surprises only the LSAP factor produces the positive medium-run gradient; the current-rate and forward-guidance factors do not, and are

[13] We use the elasticity of Saiz (2010b) as a robustness measure, which matches 217 areas. The two measures correlate about 0.4 in the common sample, and the baseline elasticity correlates about −0.13 with the young-to-old ratio.

Table 13: LSAP Age Gradient at Four Quarters: Sensitivity to the Housing-Supply Control

Specification	MSAs	Coefficient	S.E.	p-value
No supply control, full sample	366	0.054	0.023	0.020
No supply control, common sample	296	0.067	0.017	< 0.001
LSAP× BSH, common-demand	296	0.056	0.035	0.107
LSAP× BSH, perfect-substitutes	296	0.054	0.037	0.139
LSAP× Saiz	217	0.086	0.073	0.242

Notes: Each row reports the four-quarter ($h = 4$) age gradient β_4 from (6), in percentage points per standard deviation of the young-to-old ratio, under a different treatment of housing-supply elasticity. The first two rows omit the $Z_t \times \text{Supply}_m$ interaction; the "common sample" restricts the no-control estimate to the 296 metropolitan areas for which the baseline Baum-Snow and Han (2024) elasticity is available. The last three rows add the interaction, using the baseline common-demand aggregation, the perfect-substitutes aggregation, and the Saiz (2010b) elasticity, respectively. Comparing the two 296-area rows, adding the supply control leaves the point estimate almost unchanged (0.067 to 0.056) but roughly doubles the standard error (0.017 to 0.035), so significance at four quarters is lost.

sometimes negative. We read the LSAP reduced form as suggestive rather than as a stable ranking across instruments. Second, the identifying window is short. Post-2000 asset-purchase surprises deliver few independent easing episodes at the quarterly frequency, so inference is preliminary and we do not yet attach randomization-inference p-values. Third, the annual county version of the exercise loses significance once we add state-by-year effects, which indicates that part of the cross-sectional signal reflects composition across states. The quarterly metropolitan reduced form with the supply control is our preferred specification, and it is the moment the model targets.

E Tenure Choice: Capital Gains and Rent Hedging

Expectations of house-price growth raise housing demand through two channels whose strength differs across the life cycle. To isolate them we compare owning and renting for a household with n periods of remaining life in a deliberately stripped-down case: we set $\beta R = 1$ and abstract from income risk, so the tenure choice reduces to the cost comparison below, and we treat the mortgage and down payment as a financing cost that does not vary with expectations. None of this changes the sign or life-cycle incidence of the two channels we derive. The household expects house prices

to grow at rate γ per period, so a house worth p today is worth $p(1+\gamma)^k$ in k periods, and future payments are discounted at the gross interest rate R.

Let $S = \sum_{k=0}^{n-1} \left[(1+\gamma)/R\right]^k$ be the present value of one dollar paid at the start of each of the n periods. Renting costs $\chi\, p\, S$, since the rent χp grows with the price. Owning—buying now at p, paying maintenance μp each period, and selling at the end—costs $p + \mu\, p\, S - p(1+\gamma)^n/R^n$. The cost of owning relative to renting is therefore

$$G(n, \gamma) = p\left[1 - \tfrac{(1+\gamma)^n}{R^n}\right] + (\mu - \chi)\, p\, S. \tag{16}$$

A household owns when this net cost is small enough to be outweighed by the ownership premium ω; richer households own, so the marginal buyer is the income type just indifferent between the two, and a fall in G moves that margin down and raises ownership. Because ω does not depend on expectations, how tenure responds to γ is governed entirely by how G responds to γ.

Differentiating (16) at $\gamma = 0$,

$$\frac{\partial G}{\partial \gamma} = -p\Big[\underbrace{\tfrac{n}{R^n}}_{\text{(a) capital gain}} + \underbrace{(\chi - \mu)\sum_{k=1}^{n-1} \tfrac{k}{R^k}}_{\text{(b) rent hedge}}\Big] < 0, \tag{17}$$

so higher expected price growth lowers the cost of owning through two terms. The *capital-gain* channel (a) runs through the resale value and is collected once, at sale. The *rent-hedge* channel (b) runs through the rising rents that owning avoids; since $\chi > \mu$, a household that keeps renting pays a cost that grows faster than an owner's maintenance, and this accrues in every remaining renting period (Ortalo-Magne and Rady, 2006).

The two channels split by age. Capital gains (a) are collected once, at sale, so they matter most for the *old*, who have few periods left to earn them. The rent hedge (b) accrues every remaining period, so it matters most for the *young*, who face the longest horizon of rising rents, and vanishes for the last cohort. A younger age structure therefore shifts housing demand toward the rent-hedge channel and toward the households most sensitive to expected price growth—whether this raises

the aggregate response depends on how many households sit at that margin, which Section 3.6 takes up quantitatively.